Constructive phase representations

Exact counting beyond the number of intermediate states

Astra P3 research

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Abstract

Conditional on the pinned OpenAI counting manuscript, we construct an exact counter for products of fixed-locality, fixed 2-power root-of-unity phases on n independent bits, with $m = n\exp(o(\log n))$ factors, in $\mathrm{poly}(L+n)2^{o(n)}$ bit time and polynomial space. The interaction pattern is unrestricted. A common divisibility filtration makes the retained input degree $O(n)$ independently of m, while retaining one small scope per local operation. The source's replacement forms and shared summaries are then manufactured at growing incidence, rather than supplied.

The result gives exact preimage counting for checked bounded-degree arithmetic programs whose compiled outputs meet the stated support and output-count bounds, even with many intermediate states, and exact amplitudes and requested marginals for circuits with $O(n)$ Hadamards and $n^{1+o(1)}$ local diagonal phase gates. Raw descriptions are compiled, with expansion, reconstruction, and arithmetic costs charged. Failed repairs of the general-circuit problem are recorded as well. This does not establish subexponential counting for arbitrary circuits or a practical speed advantage. Executed programs validate the representation and finite identities; they do not implement the source's full asymptotic replacement solver.

1 Result, dependency, and the obstruction removed

All algorithmic claims using replacement forms are conditional on [1], specifically Definition 3.1, Proposition 3.4, Lemma 4.7, and the rational arithmetic argument of Appendix B. The supplied PDF was compared byte-for-byte with the pinned Git blob. The relevant sparse-equation reduction, sampling and inversion consequences, linear-size circuit consequence, and tensor-contraction appendix were read before formulating this extension. This report does not claim an independent verification of every theorem in the manuscript.

A direct circuit encoding counts on the original input bits plus gate bits. Many intermediate states can erase the improvement measured in the original inputs. Treating a global parity polynomial as one factor avoids gate bits but gives it a large scope. Splitting it into local monomial factors keeps scopes small, but ordinarily charges all factors to the interpolation degree. We remove this last cost using divisibility and pay explicitly for growing incidence on the original variables.

Theorem 1.1 (Local phase products)**.** *Fix integers $k, b \geq 1$. Set $q = 2^b$, $d = 2^{b-1}$, and let ζ be a primitive q-th root of unity. An input explicitly specifies n Boolean variables and m tables $f_j : \{0,1\}^{I_j} \to \mathbb{Z}/q\mathbb{Z}$, with $|I_j| \leq k$. Let L include all indices and tables. The exact algebraic integer*

$$Z = \sum_{x \in \{0,1\}^n} \zeta^{\sum_{j=1}^{m} f_j(x|_{I_j})}$$

can be computed deterministically in $\mathrm{poly}_{k,b}(L+n)2^{o(n)}$ bit operations and polynomial space whenever $\log(2+m/n) = o(\log n)$. No graph-width, geometry, low-rank, or supplied-decomposition hypothesis is imposed.

A polynomial exponent may depend on fixed k, b, but not on input length or density. For $n \geq 4$, set

$$\lambda = \lceil \log_2 n \rceil, \qquad \nu = \lceil \log_2 \lambda \rceil,$$
$$h = \lceil \log_2(2 + km/n) \rceil + 4\nu + \lceil \log_2(k+2) \rceil + 10, \qquad a = h/\lambda.$$

For sufficiently small a the quantitative bound is

$$(L+n)^{O_{k,b}(1)} 2^{O_{k,b}\left(n\sqrt{a}\,[\log_2(2/a)]^{3/2}\right)}. \tag{1}$$

One guarded algorithm handles all inputs, using enumeration outside the useful range; Section 3 makes this explicit.

A strict regime beyond direct gate encoding. For $0 < \alpha < 1$, let $m = n2^{(\log_2 n)^\alpha}$. The exponent becomes

$$O_{k,b,\alpha}\left(n \frac{(\log\log n)^{3/2}}{(\log n)^{(1-\alpha)/2}}\right) = o(n).$$

A direct bounded-fan-in accumulation of m local terms has $\Theta(m)$ gate states. Substituting that variable count into the governing circuit-to-CNF bound gives no improvement over 2^n in this regime. This compares the explicit reductions, not every conceivable encoding. The new bound removes a state-count and degree obstruction; it is not just the linear-size circuit corollary. Single quadratic $\mathbb{F}_2$ phase sums already have polynomial-time algorithms [2]; the substantive binary application begins at degree three. Multiple quadratic outputs are also relevant.

2 A degree budget obtained from divisibility

For $p(x) = \bigoplus_{S \in \mathcal{M}} x^S$ over $\mathbb{F}_2$, with $|S| \leq k$, form

$$F(X,V) = \prod_{S \in \mathcal{M}} (1 - 2VX^S) \quad \text{in } \mathbb{Q}[V]/(V^{n+2}). \tag{2}$$

V is a formal bookkeeping variable, not a Boolean state. Give each factor independent incidence indeterminates before measuring degree. A retained term of V-degree j selects j nonconstant factor parts, so its incidence degree is at most kj. The full retained product has degree at most $k(n+1)$, independently of m. This satisfies the exact degree hypothesis of Proposition 3.4 without globally identifying and multilinearizing all copies.

On Boolean inputs the untruncated product at $V = 1$ is $(-1)^{p(x)}$. Discarded orders $j \geq n+2$ are divisible by 2^{n+2}. After reconstructing the retained polynomial at the undeformed point,

$$\sum_x \sum_{j=0}^{n+1} [V^j] F(x,V) \equiv Z_p := \sum_x (-1)^{p(x)} \pmod{2^{n+2}}.$$

Since $Z_p \in [-2^n, 2^n]$, centered reconstruction recovers it and $\#\{x : p(x) = 0\} = (2^n + Z_p)/2$. Just $n+1$ modular bits would identify the endpoints $+2^n$ and -2^n.

For general phase tables, multilinearly interpolate $x \mapsto \zeta^{f_j(x)}$ to $G_j \in \mathbb{Z}[\zeta][X_{I_j}]$, of degree at most k and with at most 2^k coefficients. This costs $O_k(1)$ operations per explicit table. Put $\pi = \zeta - 1$. Modulo (π) each table is identically one. Boolean finite differences therefore show that every coefficient of $G_j - 1$ is divisible by π.

Lemma 2.1 (Exact finite filtration)**.** *Set $P = n+2$, $\ell = dP$, and*

$$\mathcal{A} = \mathbb{Q}(\zeta)[V]/(V^\ell), \qquad B_j = 1 + V(G_j - 1).$$

The retained product $\prod_j B_j$ has incidence degree at most $K = k(\ell - 1)$. Its coefficient sum on Boolean inputs agrees with the original phase product modulo $2^P\mathbb{Z}[\zeta]$.

Proof. Count the selected nonconstant parts for the degree bound. Also $\pi^d \in 2\mathbb{Z}[\zeta]$. For $d = 1$, $\pi = -2$. For $d \geq 2$, expand $(\zeta - 1)^d$: the endpoints cancel since $\zeta^d = -1$, and all intermediate binomial coefficients are even because d is a power of two. An omitted term has at least dP factors in (π) and hence belongs to $2^P\mathbb{Z}[\zeta]$. □

Use the integral basis $1, \zeta, \ldots, \zeta^{d-1}$, with $\zeta^d = -1$. Each input contributes one signed basis vector, so every coordinate of Z lies in $[-2^n, 2^n]$. Its residues modulo 2^{n+2} determine all coordinates exactly. No ramification unit is needed.

Remark 2.2 (A final linear map, not a homomorphism). Setting $V = 1$ is not an algebra homomorphism on the truncated ring. Sum coefficients in its unique representative of degree below ℓ only after exact reconstruction at the undeformed point. The lemma justifies this final map. The modular congruence is required only at original Boolean inputs; the non-Boolean, rational replacements at other points are handled by exact rational arithmetic.

3 Manufacturing the replacement forms

Add a neutral factor of scope $\{v\}$ for each variable, so total incidence $M \leq km + n$. Branch on variables belonging to more than

$$B_0 = \lceil M/n \rceil \lceil \sqrt{\lambda} \rceil$$

factor scopes. There are at most $n/\sqrt{\lambda}$ such variables. Branches partition the input cube and leave $N \geq n/2$ free bits for sufficiently large n. Substitute fixed bits in the local tables and retain neutral factors. The maximum remaining incidence B is at most $B_0 + 1$, the scope bound w is at most $\max(k, 1)$, and $\log_2 B = O_k(h)$.

Put $\lambda' = \lceil \log_2 N \rceil$ and recompute a comparable h' from the actual branch, retaining original precision $P = n + 2$. Choose

$$r = \lfloor \sqrt{\lambda'/h'} \rfloor, \qquad t = \lceil \log_2 r \rceil, \qquad Q = \lfloor \lambda'/(128t) \rfloor,$$
$$\tau = (r\lambda')^{-1}, \qquad D = r^{2Q}, \qquad A = \lceil \log_2(2Br^2\lambda') \rceil,$$
$$R = \lceil \sqrt{8QA} \rceil + 32.$$

Check $r \geq 2$, $R \geq 32$, $6R \leq Q$, central-interval endpoints, and the source's two balance conditions

$$2(B + w)D^2 \leq M_{\text{branch}}, \qquad 2Qr^2 \leq N.$$

As $a \to 0$,

$$A = O_k(h), \quad D \leq (2N)^{1/64}, \quad B = N^{o(1)}, \quad R/Q = O_k(\sqrt{a \log(2/a)}).$$

The balance guards thus hold eventually. An R-point central interval contains the integers within $R/4$ of the source binomial mean. Its exceptional fraction is bounded by

$$2\exp(-R^2/(8Q)) \leq 2^{1-A} \leq (Br^2\lambda')^{-1} = \tau/(rB).$$

Lemma 4.7 now constructs the occurrence-specific replacement rows, partitions, and shared integer measurement matrices. Rows restore original coordinates at zero and have degree at most R. Groups have size at most $2N/r$. There are at most $2\tau N$ measurement rows in total, each with absolute coefficient sum at most D. All are manufactured in polynomial time in their explicit dimensions.

Uniform fallback. R/Q need not tend to zero at a fixed positive a; this cannot be hidden in a bounded-exceptions statement. Below a sufficiently small fixed threshold for a, the guards hold uniformly for sufficiently large n. Above it use original-input enumeration. Its 2^n cost is absorbed by increasing the constant in (1) on a bounded positive-a interval; larger a simply uses the stated fallback. Thus one algorithm has $2^{o(n)}$ cost along every family with $a \to 0$.

4 Summation, reconstruction, and bit costs

Apply Proposition 3.4 in the commutative rational algebra $\mathcal{A}$, of dimension $d\ell = O_b(n)$. The degree K is $O_{k,b}(N)$ since $N \geq n/2$. The Taylor correction order is

$$q_T = \lfloor KR/Q \rfloor = O_{k,b}(n\sqrt{a\log(2/a)}).$$

For $q_T \leq N/2$, the subset-pair enumeration satisfies

$$C_{q_T} = \sum_{j \leq q_T} 2^j \binom{N}{j}, \qquad \log_2 C_{q_T} \leq N(H_2(q_T/N) + q_T/N).$$

This supplies the entropy logarithm in (1). Measurement and group enumeration cost

$$(2D+1)^{2\tau N} 2^{2N/r} = 2^{O(N/r)} = 2^{O(n\sqrt{a})}.$$

For other correction orders use $C_{q_T} \leq 3^N$, absorbed where the claimed exponent is already a constant multiple of n. The high-incidence branches add $n/\sqrt{\lambda}$, also absorbed.

Let $\delta = (R-1)!Q!$ be the source's row denominator. A retained V^j coordinate uses at most kj replacement inputs, so its denominator divides δ^{kj}, including in Taylor evaluation. After clearing denominators a coefficient-norm bound is a sum of terms of the form

$$\binom{m}{j} (C_{k,b} \max(\delta, B_*)^k)^j, \quad j < \ell, \qquad \log B_* = O(Q\log(Q+2) + \log(nD+2)).$$

Its logarithm is polynomial in $L+n$ and the design parameters. Summing exponential enumerations adds their logarithm to bit length. Cyclotomic multiplication uses the fixed relation $\zeta^d = -1$. Truncated Taylor/V arrays and Hermite systems have polynomial dimension, and rational reconstruction has polynomial bit cost by the Appendix B argument coordinatewise. Reconstruct at $T = 0$, sum retained V coefficients, reduce integral coordinates modulo 2^{n+2}, and center. Sequential traversal gives polynomial space. This proves Theorem 1.1.

5 Compiling raw arithmetic programs and preserving outputs

Consider an explicitly supplied program over $\mathbb{F}_2$ or $\mathbb{Z}/2^b\mathbb{Z}$ with n Boolean inputs and S arithmetic gates. For fixed formal degree k, propagate canonical Boolean multilinear polynomials through the actual gate DAG. Addition combines coefficient dictionaries; multiplication takes pairwise unions of supports, using $x_i^2 = x_i$, and combines coefficients modulo the base ring. Each stored polynomial has at most $\sum_{j \leq k} \binom{n}{j} = O_k(n^k)$ terms. A conservative expansion cost is

$$O_{k,b}(Sn^{2k})$$

coefficient operations, with index-bit costs polynomial in the explicit description. Formal degree is checked syntactically before claiming this bound. The supplied Boolean compiler also accepts the broader class where every intermediate canonical polynomial has degree at most k: it fully forms each product of two degree-k dictionaries, then checks the actual degree before retaining it. A degree violation is reported, not silently truncated. This uses the same conservative polynomial bound.

The final support is manufactured by this computation. An arbitrary program merely promised to have a sparse final expansion is not assumed to have an efficient compiler. If the constructed support is too large, the favorable counting bound is not claimed. The result therefore provides a checkable representation compiler for a restricted algebraic class, rather than a universal compression assertion.

For a single output modulo q, compute all q character sums and then Fourier-invert after their exact reconstruction. One primitive character alone need not determine a q-value histogram when $q > 2$.

Corollary 5.1 (Vector target-preimage counting)**.** *Suppose the compiler obtains $t \le Cn$ output polynomials P_i of degree at most fixed k, with total nonzero support $m = n\exp(o(\log n))$. For a supplied target y, its exact original-input preimage count is computable in compilation cost plus $\mathrm{poly}(L+n)2^{o(n)}$ time and polynomial space. There is no charge of one counting variable per intermediate program state.*

Proof. Introduce bt character bits encoding $\lambda_i \in \mathbb{Z}/q\mathbb{Z}$. Orthogonality gives the exact integer identity

$$\sum_{x,\lambda} \zeta^{\sum_{i=1}^{t} \lambda_i (P_i(x)-y_i)} = q^t \, \#\{x : P_i(x) = y_i \text{ for all } i\}.$$

Expand λ_i into its b bits. The phase now has $N = n + bt = O(n)$ independent bits, locality at most $k+1$, and at most $bm + bt$ monomial gates. Theorem 1.1 applies. Precision is $N+2$, not $n+2$, so the multiplicity is fully charged. Reconstruct the character sum exactly, then divide by q^t exactly. □

For a raw local accumulation program the local factors are already explicit instructions, so this polynomial-expansion stage is unnecessary. Its arbitrarily long sequence of prefix states is replaced directly by the product construction. In both formulations the resulting global output may depend on all n inputs; no constant output-locality hypothesis is used.

6 Exact quantum amplitudes and requested marginals

6.1 IQP circuits

For $U = H^{\otimes n} D H^{\otimes n}$, with D a product of the permitted local diagonal phase gates, the amplitude at y is

$$\langle y|U|0^n\rangle = 2^{-n} \sum_{x} \zeta^{P(x)} (-1)^{x\cdot y}.$$

The extra phases are unary. For output positions A, $|A| = a_0$, and $B = [n] \setminus A$, summing the unspecified output characters gives

$$\Pr[Y_A = z] = 2^{-(n+a_0)} \sum_{x_B, x_A, x'_A} \zeta^{P(x_A,x_B) - P(x'_A,x_B)} (-1)^{z\cdot(x_A + x'_A)}. \tag{3}$$

Indeed orthogonality forces the two B-input copies to agree. This is a phase sum on at most $2n$ bits with at most $2m + 2a_0$ local factors, so Theorem 1.1 computes it exactly. No list of all 2^n amplitudes is produced for free.

6.2 Arbitrary alternation with a linear Hadamard count

The following extension was constructed and checked separately from the IQP case. The path expansion itself is established machinery [6]; the new complexity bound comes from the phase counter.

Corollary 6.1. *Fix k, b, C. Let a raw n-qubit circuit, on computational input $|0^n\rangle$, have $h_H \le Cn$ Hadamard gates and $m = n\exp(o(\log n))$ diagonal gates of locality at most k, with phases in the 2^b-th roots of unity. The gates may alternate arbitrarily and have arbitrary interaction geometry. Every requested amplitude and every requested computational output marginal is exactly computable in $\mathrm{poly}(L+n)2^{o(n)}$ bit time and polynomial space, conditional on the governing manuscript.*

Proof. Track the computational bit on each wire. A diagonal gate leaves this bit unchanged and contributes a local phase on the current bits. At a Hadamard with old bit u, introduce one new path bit v and use

$$\langle v|H|u\rangle = 2^{-1/2}(-1)^{uv}.$$

Thus at most h_H path bits are introduced, with no new bit at any diagonal gate. Initial wire values are constants. The accumulated phase has at most $m + h_H$ local terms of locality at most $\max(k, 2)$.

For an amplitude, pin the final wire values to the requested output. These are pins on path bits, or tests of constants. Contradictions give zero. Eliminate consistent pins by substitution. Each remaining path assignment corresponds to exactly one compatible computational history. The amplitude is $2^{-h_H/2}$ times its phase sum.

For a marginal, take a path expression and its complex-conjugate copy. Pin measured final wires in both copies. Identify the two final bits on each unmeasured wire, implementing the sum over the unspecified output. These equalities and pins are constructed by union-find. Their elimination does not enlarge any local scope or introduce multiplicity. There are at most $2h_H$ free variables and $2(m + h_H)$ local terms. The probability is 2^{-h_H} times the resulting phase sum.

To keep a uniform parameter comparable to the physical qubit count, pad either expression with unused Boolean variables to $N_* = n + 2h_H = \Theta(n)$. This multiplies the sum by a known power of two. Use precision $N_* + 2$, reconstruct exactly, and divide by this known power after reconstruction. The factor density is $N_*^{o(1)}$. Theorem 1.1 applies. Return amplitudes as exact cyclotomic coordinates times $2^{-h_H/2}$; if necessary $\sqrt{2}$ is adjoined to the fixed number field. All normalization and boundary processing have polynomial bit cost. □

The compiler and an independent state-vector evaluator agree exactly on the finite tests in Section 9. The restriction on Hadamard count is substantive. Arbitrary gate sets cannot be converted without charging any added Hadamards. Exact sampling is a standard conditional-probability consequence in this extended family: fixed-degree algebraic probabilities admit certified dyadic comparisons, and a random dyadic threshold is resolved in expected polynomial work. No fast sampler or practical simulation advantage is claimed to have been executed here.

7 Why the direct general-circuit attacks failed

The research did not stop at the first special case. The following repairs were constructed, tested, and rejected at their precise scope. Full proofs and exact checkers are included in the accompanying archive.

7.1 Consistency tests do not eliminate arbitrary trace entropy

For a topologically ordered Boolean circuit with inputs x and stored gate bits $y_1, \dots, y_s$, define residuals

$$e_g = y_g + f_g(x, y_1, \dots, y_{g-1}) \quad \text{over } \mathbb{F}_2.$$

The map $(x, y) \mapsto (x, e)$ is bijective: recover y_g successively. Even at fixed x, every one of the 2^s residual vectors occurs. A rank-r parity filter on residuals therefore leaves exactly 2^{n+s-r} traces. Unlike the source's clause-violation vectors on the original input cube, these residuals cannot all be removed by a shorter exact linear consistency filter.

This is not a lower bound against output-specific algorithms. We tested that escape too. Write $\chi(e) = (-1)^{e_1+\cdots+e_s}$. If a rational weight w preserves both the constant observable and the pure parity observable at $e = 0$, then $\sum_e w(e)\chi(e) = 1$. Every polynomial of degree less than s is orthogonal to χ, since each monomial omits a coordinate. Thus such a selector has degree at least s, even with signed weights. Alternatively a rank-one parity filter can preserve this particular output, but its multiplicity is 2^{s-1}. More generally an invariant rank-r filter gives $Z_{\text{filt}} = 2^{s-r}Z$; recovering arbitrary $0 \le Z \le 2^n$ from a power-of-two residue requires at least $n + s - r + 1$ bits before division. A small filter alone does not remove the state cost. The new output-character construction avoids this failure by using only $O(n)$ character bits and charging their precision.

7.2 A shared linear bottleneck has a degree budget

If an $n \times n$ polynomial matrix $P(T)$ has entry degree at most R and $P(0) = I$, then at distinct nodes $t_1, \dots, t_Q$,

$$\sum_j (n - \operatorname{rank} P(t_j)) \le nR.$$

Indeed, the rank deficiency at t_j is a lower bound on the order of vanishing of $\det P$ there, while $\det P$ is nonzero and has degree at most nR. In particular, a common rank-$o(n)$ image at every cheap node forces $R/Q \ge 1 - o(1)$. Incidence-specific forms are essential; one cannot simply replace them by one small universal input summary.

The stronger signed-alias repair also failed for the tested complete auxiliary grid. For $n = r^Q$, the source operator $M = Vf(W)$ with $\deg f \le R - 1$, and every nonzero rational signing/scaling J_w of replicated inputs, a valid summary satisfies

$$\operatorname{rank}(CJ_w) \ge \sum_{j=R}^{Q} \binom{Q}{j} (r-1)^j.$$

At $R/Q = o(1)$ this is almost all original-input rank. The proof uses the first exceptional spectral level and triangular equations on swap cubes; the unsigned bound is exact. This rules out that repair of the source grid, not all possible representations.

7.3 Causality, precision, and the tensor bridge

Imposing triangularity on the source replacement operator has an immediate cost: every forced zero diagonal entry of M contributes a pivot to $I - M$. If there are b_* such entries, any summary retaining its fixed space has rank at least b_*. In a finite 64-coordinate fixture the noncausal source operator has defect rank 10, whereas any triangular operator satisfying the same support restriction has defect rank at least 32. Repair by deleting variables to make the auxiliary dependency graph acyclic also consumes almost all variables in the tested regime.

The Appendix D interpolation precision cannot simply be replaced by the output bit length. Explicit integer polynomial families have value one at the target but vanish to precision $\Omega(MD/u)$ at all active local evaluation nodes. A separate pinned-wire fixture exhibits the same issue at small size. The present phase construction instead truncates a globally justified polynomial *before* rational interpolation; it does not assume that tensor precision floor disappears.

N1077 was tested at its actual generic planar trivalent rational scope [7]. The tensor proof and packet's gauge admissibility conditions were read. An AND-chain edge has zero cross-pairings in the relevant canonical decomposition, and a separate nondegenerate rational theta network fails the required compatibility products. A simple low-degree deformation cannot supply arbitrarily many compatible nodes without paying the cleared compatibility degree. These checks do not refute N1077; they show why it does not supply a generic circuit contraction bridge for this task. No such bridge is used in the positive theorem.

8 Other completed bounds and prior-art comparison

A separate completed hybrid compiler treats a 2-CNF B together with direct clauses C having H literal incidences. It manufactures the source parity summary for B alone and attaches physical variable ports for C, obtaining $O(N \log\log N + H)$ Boolean wires. The key invariant is equality of the solution sets, not just equality of scalar counts. Combining this with a disjoint long-clause split gives raw AND/OR circuits with arbitrary fan-in and g gates the bound, for $N = n + g$,

$$(L+N)^{O(1)} 2^{O\left(N\sqrt{\log\log N}/(\log N)^{1/4}\right)}.$$

It permits dense wiring, but still pays for the gate variables. The qualitative $g = O(n)$ consequence is already obtainable from fixed width and a simple split, so it is not the principal advance claimed here. An earlier uniform growing-width theorem and its independent audits are also retained in the archive.

The targeted primary-source comparison supports a substantive conditional extension, not a first-ever priority claim. Williams [2] separates exact constant-degree polynomial counting from approximate quadratic-system counting. Complex weighted counting dichotomies [3] provide relevant hardness and tractable classes, but not a reverse original-variable-preserving encoding eliminating arbitrary factor density. Codsi–van de Wetering [4] exploit random graph structure for exact IQP improvements; their sparse-instance bound gives savings from 2^n, rather than this worst-case $2^{o(n)}$ guarantee. Wahl–Strelchuk [5] already give a genuine subexponential simulation result under finite-range fixed-dimensional geometry. De Colnet et al. [6] construct path representations and evaluate them through rank-width. Small rank-width can cover instances outside our density range, so neither method uniformly dominates the other.

A selective search of the pinned OpenAI release catalog checked the entries on noncommutative formula identity testing, exact Fourier transforms, finite-monoid expressions, one-tape space simulation, and the governing counting result. Their stated contracts did not supply a ready-made time-efficient generic-program compression step. This is a search finding, not an impossibility claim or an exhaustive archive review.

9 Executed programs and preserved sources

All new finite validation reported here used exact integer or rational arithmetic and passed. Distinct checkers were written independently. The rows below count their actual executed fixtures; they are not timing benchmarks or executions of the full asymptotic solver.

Executed component	Verified scope
Principal phase construction	30 truncation cases at orders $2, 4, 8$; 18 raw arithmetic programs; 6 multiple-output reductions; 168 IQP prefix identities.
Independent phase audit	72 fixtures; 336 character reconstructions; insufficient-precision and single-character-histogram counterexamples.
Root Boolean program compiler	21 raw programs; 1,524 input assignments; 84 exact target-preimage counts; 9 cyclotomic phase cases; 93 IQP prefix identities; explicit rejection of a degree violation.
General H/diagonal path compiler	24 circuits, including arbitrary local phase tables; 144 amplitude and 264 marginal comparisons against exact state vectors; 2 inconsistent boundaries and 2 constant-wire table checks.
Dense-grid obstruction	29 exact spectral-rank certificates and 120 signed-weight cases.
Hybrid binary/direct-clause compiler	57 fixtures; 9,561 incidence subsets; 38,244 subset-index round trips; 1,192 boundary-moment identities.

The finite counters use exhaustive assignments, exact subset-union dynamic programming, or exhaustive finite contractions. Their purpose is to validate compilation, truncation, observables, and reconstruction. There is no practical speed claim and therefore no claim of beating equally informed deployed competitors. Implementing and optimizing the complete replacement solver remains separate engineering work.

The supplied latest passive-network source, its full scientific audit, and its receipt are preserved unchanged in the archive. Their reported 290/932-reading experiments are producer reports,

not replayed network experiments from this turn. The receipt's distinction between conditional matrix concavity, derivative bounds, and physically acquired information is retained. The older 510-reaction exact/noise-aware rank-one Gram source is referenced by these documents, but its original proof and program files were not recovered through the available selective routes. No claim of preserving or replaying those missing bytes is made.

10 Scientific boundary and compact continuation

The established extension is a reusable representation principle: *bounded final coefficient height plus divisible local increments can make reconstruction degree depend on original-input precision, rather than on the number of operations.* It applies here because fixed 2-power phases have both properties, and the manufactured summary cost tolerates incidence $n^{o(1)}$.

It does not cover arbitrary dense cubic polynomials with $\Theta(n^3)$ terms, unrestricted arithmetic degree, unrestricted Hadamard count, arbitrary rational weights, or generic tensor networks. For density $m = n^{1+\varepsilon}$ with fixed positive ε, the controlling ratio $h/\log n$ does not vanish. Generic output support need not be small or cheaply certifiable. The proof is conditional on the governing manuscript, internally audited here, and has not received external peer review.

The next substantive frontier is now exact: remove the growing-incidence cost for dense cubic phases or for dense affine/quadratic preprocessing without converting it into a large scope or many independent trace bits. The signed complete-grid shortcut and output-normalization shortcut have explicit obstructions in the accompanying proofs. A continuation should start from the phase filtration and its exact bit costs, not from those rejected shortcuts or from the earlier dense-CNF corollary. All actual proofs and executed sources needed to resume are included; this statement does not imply computation continues after the response ends.

References

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